Final project manuscript

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Final project manuscript before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Independent reproduction check

Without opening the old lesson, explain one key idea from Independent reproduction check. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Writing results and limitations

Without opening the old lesson, explain one key idea from Writing results and limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Negative results and claim revision

Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Final project manuscript trace and record the first step you can explain without help.

Use W2: Trace the worked case for Final project manuscript. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Final project manuscript explanation against the supplied facts before discussing it.

Use the worked case for Final project manuscript. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Manuscript says12 evaluated cases, raw record has10 completed and2 missing with8 passes. Correct the summary.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Final project manuscript facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Final project manuscript facts and keep your reasoning.

Abstract says100% success; results table shows18/20, and two failures are omitted from the figure. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Final project manuscript facts and keep your reasoning.

A report omits this qualification: A manuscript is ready only when its claims and artifacts align; formatting alone cannot certify **evidence**.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Final project manuscript facts and keep your reasoning.

Manuscript says12 evaluated cases, raw record has10 completed and2 missing with8 passes. Correct the summary.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Final project manuscript facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Final project manuscript to Reproducible AI Research Study, identify a limit and choose the next useful test.

For Reproducible AI Research Study, use this week's independent task as a rehearsal for the final **evidence** pack: Manuscript says12 evaluated cases, raw record has10 completed and2 missing with8 passes. Correct the summary.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Final project manuscript

> Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases. Explain the labelled step before checking the changed condition.

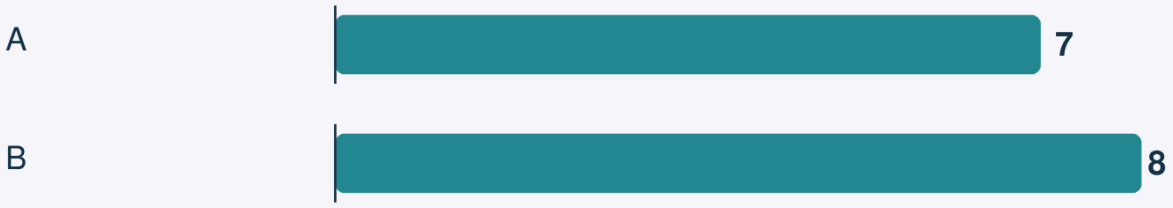
Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases.

A is correct on 7/10; B on 8/10. All cases are indoor, from five plants with two views each.

AI ACADEMY / AI-7/34

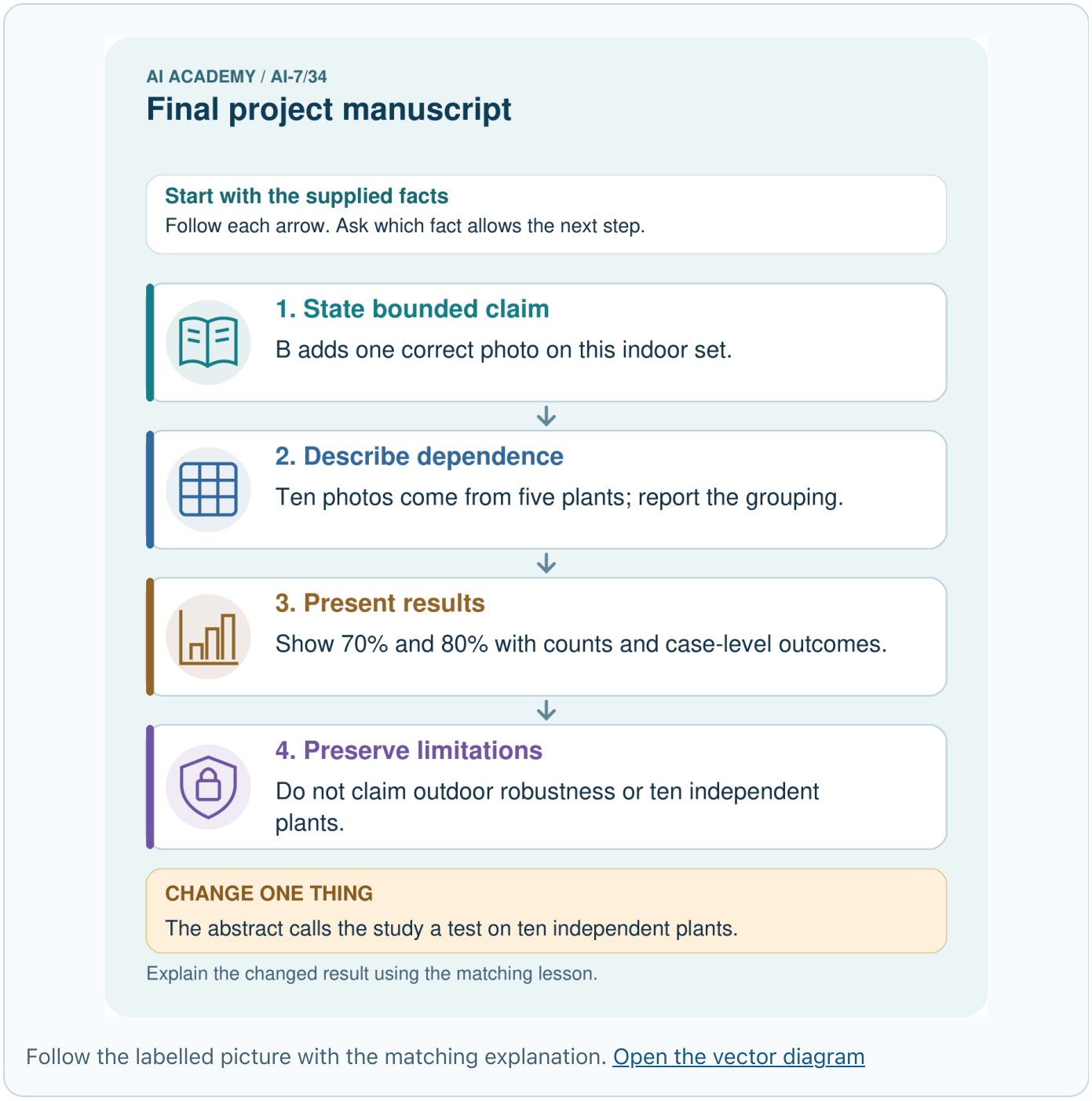
Indoor capstone photo outcomes

correct photos out of 10; five plants



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Connect the manuscript's 7/10 and 8/10 scores to the five-plant grouping. Which abstract claim must be corrected if it says ten independent plants?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Do not claim outdoor robustness or ten independent plants.

1. Present results

2. Describe dependence

3. State bounded claim

4. Preserve limitations

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Preserve limitations Do not claim outdoor robustness or ten independent plants. A final manuscript aligns question, procedure, **evidence** and scope.

Every figure and conclusion should point back to the records that support it.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The abstract calls the study a test on ten independent plants. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. Correct it to five plants with two views each.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Correct it to five plants with two views each. The source-object count, not the photo count, defines the independent plants.

Repair a missing building block

Review Independent reproduction check (ai-7/33). Goal: Explain the **mechanism** and terminology of independent reproduction check. Short practice: Locate the Q3 scoring disagreement between 5/6 and 4/6.

What should the teams inspect before choosing a score, and what limitation follows if the saved Q3 file is missing?. Return to this week and explain what you changed.

Review Writing results and limitations (ai-7/31). Goal: Explain the **mechanism** and terminology of writing results and limitations. Short practice: Compute the supported-answer rates for 9/12 and 10/12.

What actual outcome was measured, and why is a claim that every student becomes smarter outside this **evidence**?. Return to this week and explain what you changed.

Review Negative results and claim revision (ai-7/28). Goal: Explain the **mechanism** and terminology of negative results and claim revision. Short practice: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

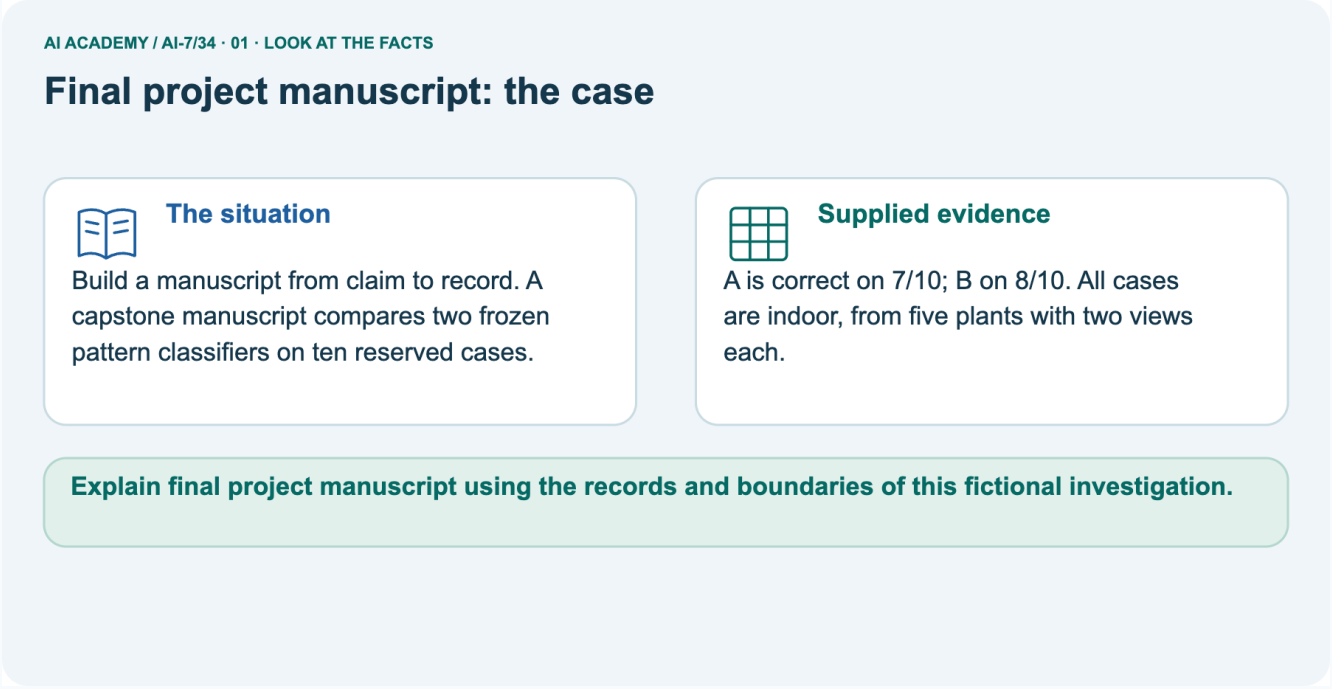
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Final project manuscript picture from its labels, then record a relevant limit.

01. Final project manuscript: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

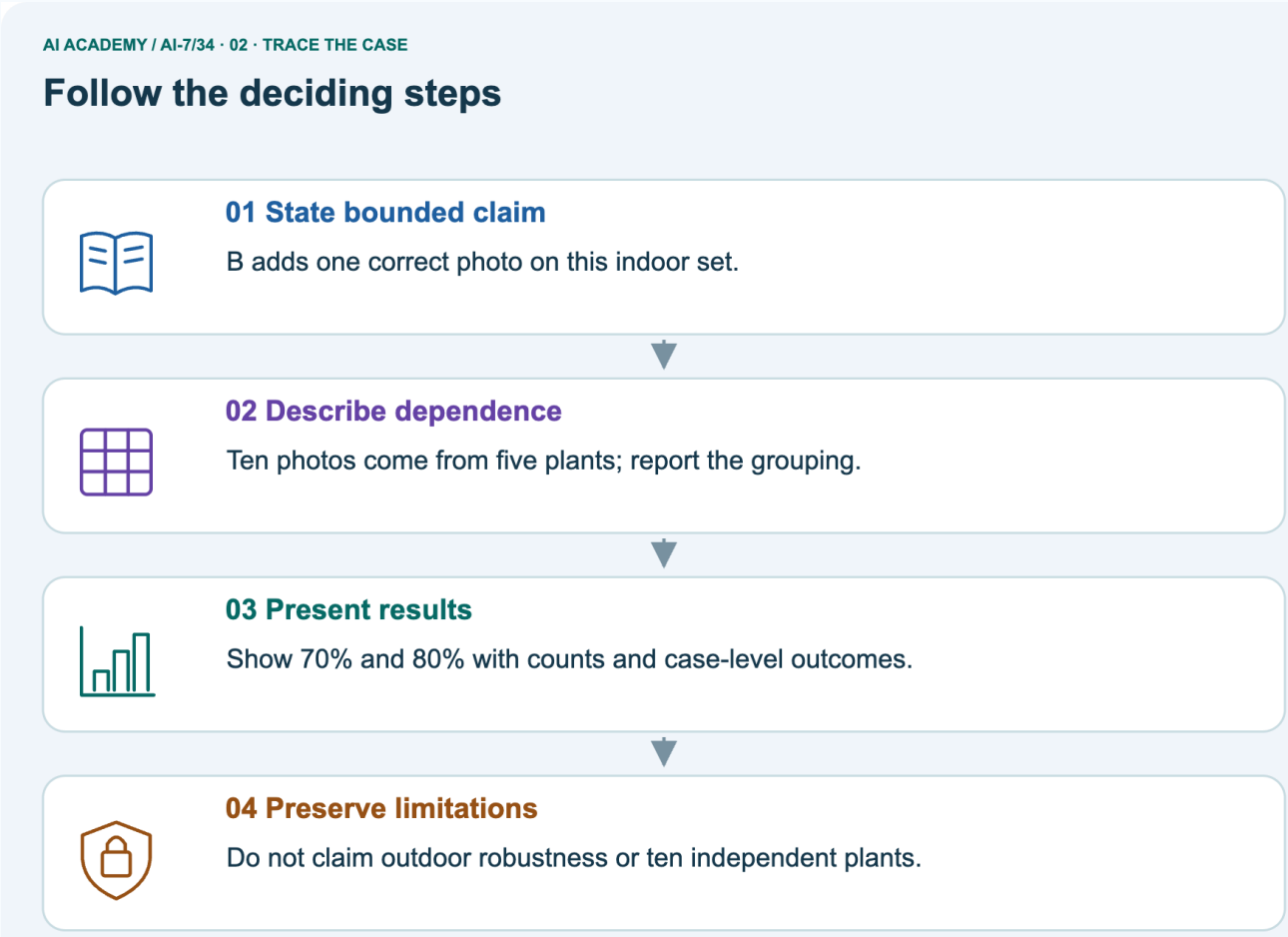
Build a manuscript from claim to record. A capstone manuscript compares two frozen pattern classifiers on ten reserved cases. Supplied facts: A is correct on 7/10; B on 8/10.

All cases are indoor, from five plants with two views each.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



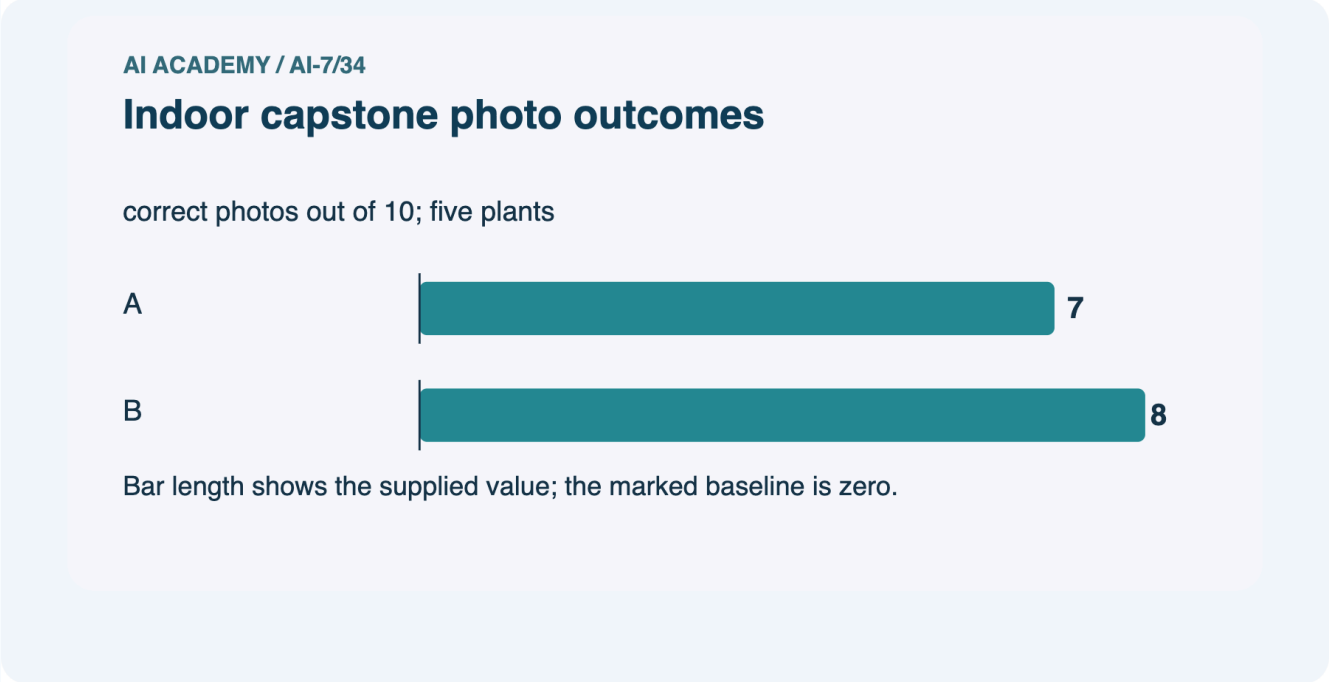
Follow the labelled picture with the matching explanation. Open the vector diagram

State bounded claim: B adds one correct photo on this indoor set. → Describe dependence: Ten photos come from five plants; report the grouping. → Present results: Show **70%** and **80%** with counts and case-level outcomes. → Preserve limitations: Do not claim outdoor robustness or ten independent plants.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Indoor capstone photo outcomes



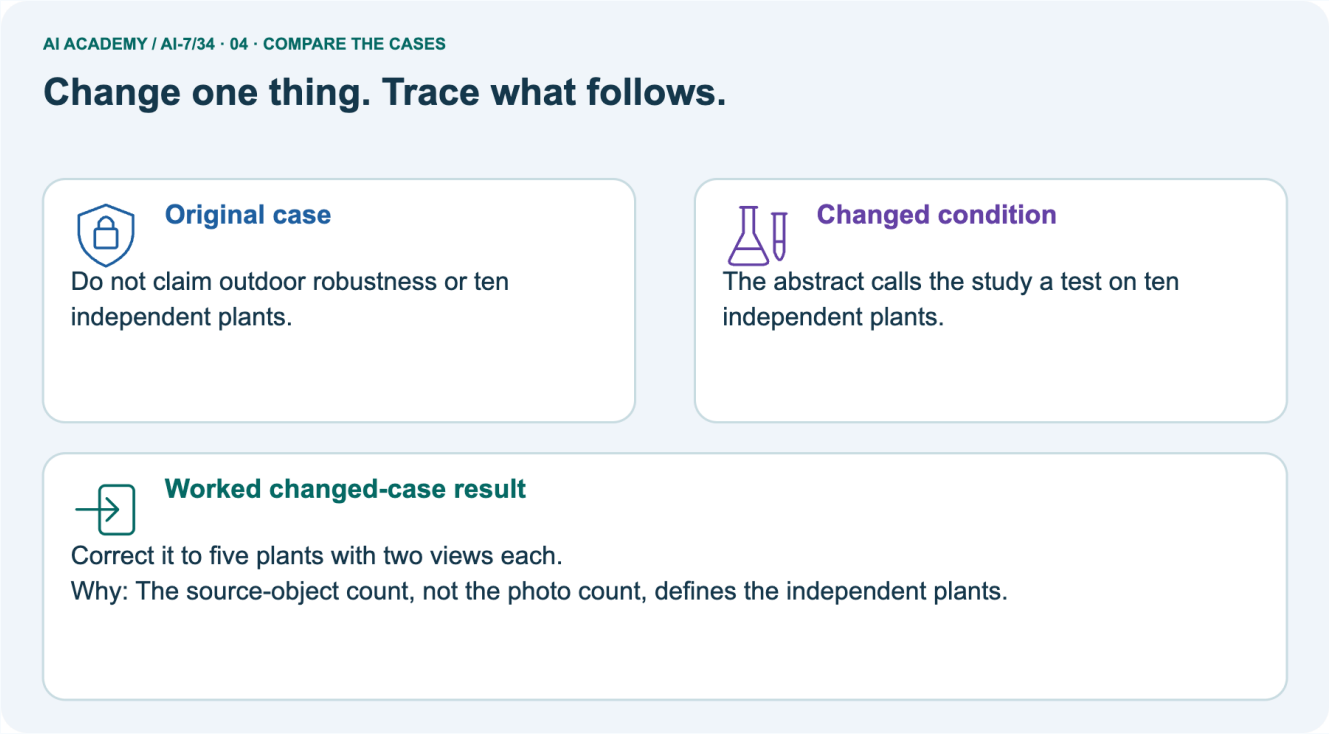
Follow the labelled picture with the matching explanation. Open the vector diagram

Indoor capstone photo outcomes. Units: correct photos out of 10; five plants. A: 7. B: 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

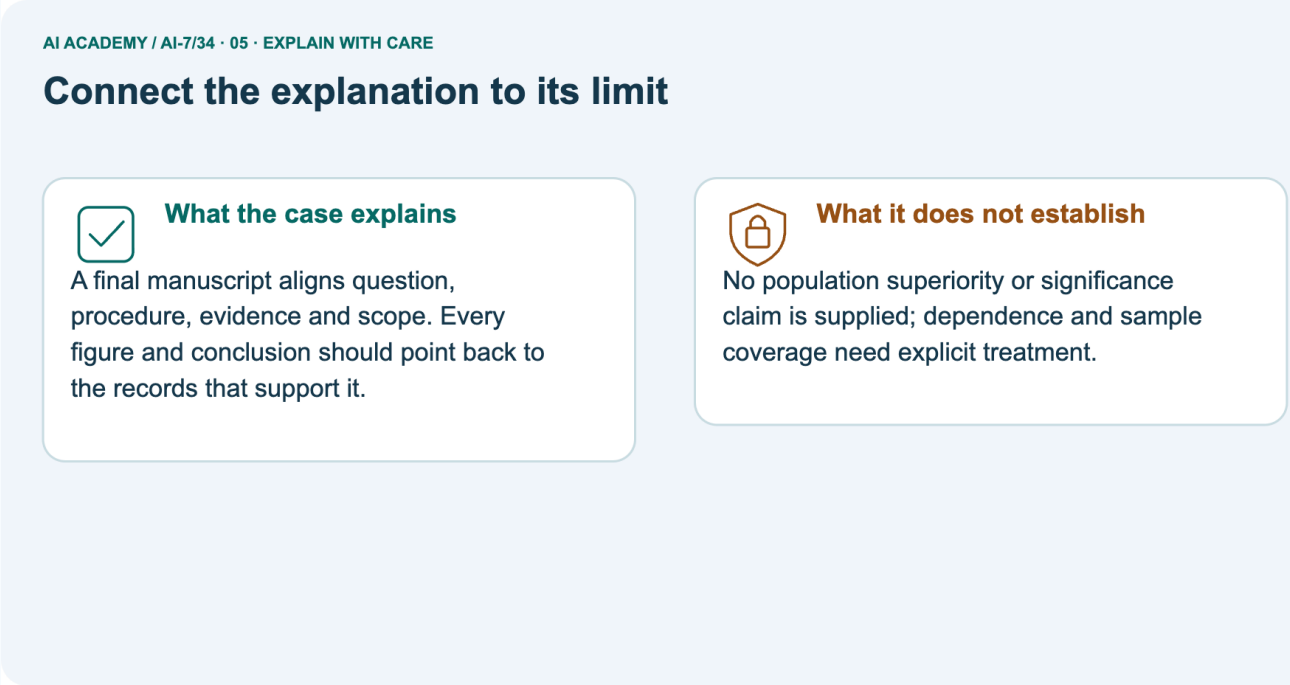
Original case: Do not claim outdoor robustness or ten independent plants. Changed condition: The abstract calls the study a test on ten independent plants. Expected result: Correct it to five plants with two views each.

Reason: The source-object count, not the photo count, defines the independent plants.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

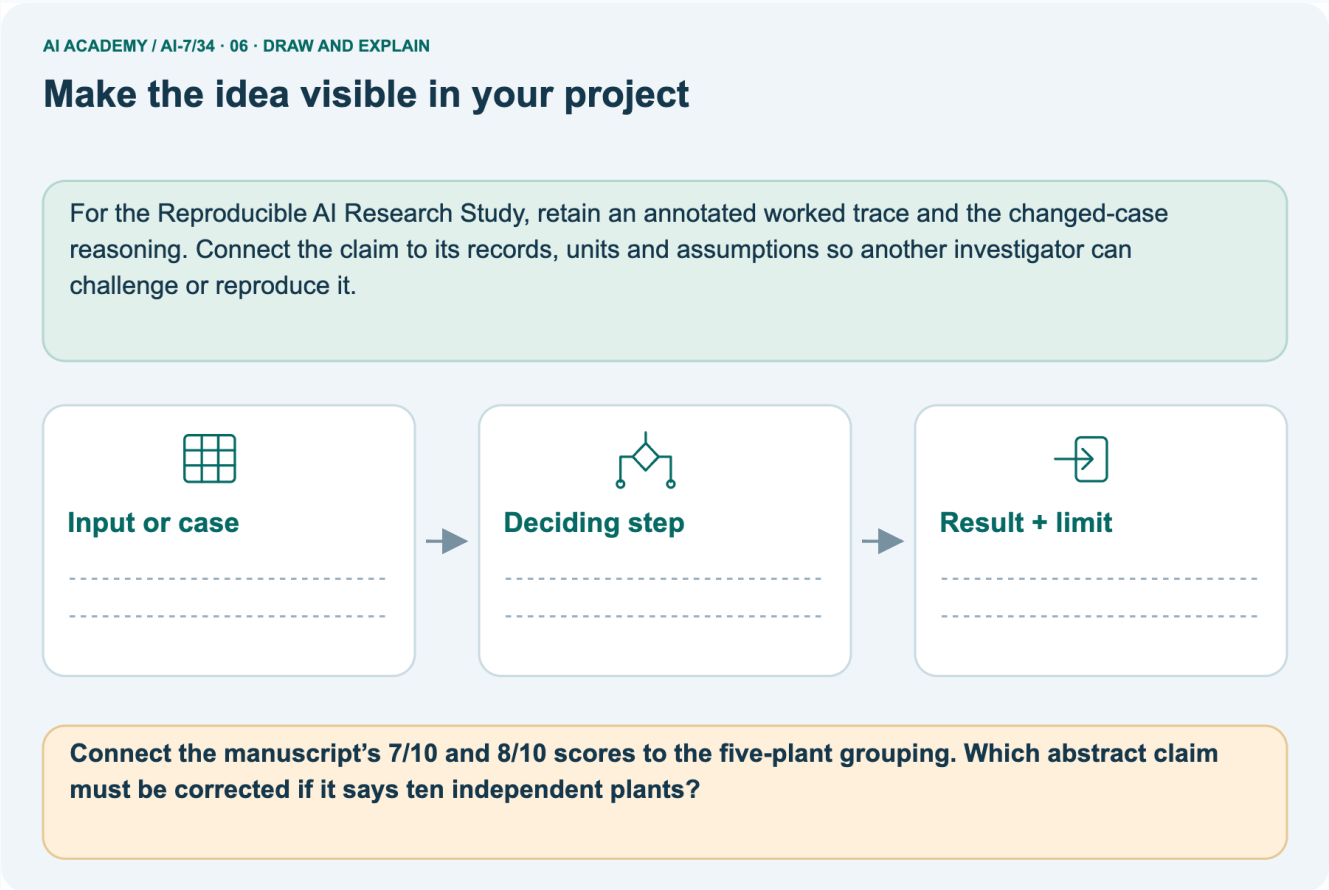
A final manuscript aligns question, procedure, **evidence** and scope. Every figure and conclusion should point back to the records that support it.

Limit: No population superiority or significance claim is supplied; dependence and sample coverage need explicit treatment.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Connect the manuscript's 7/10 and 8/10 scores to the five-plant grouping. Which abstract claim must be corrected if it says ten independent plants?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.